

Q1 - 24 January - Shift 1

If two charges q_1 and q_2 are separated with distance 'd' and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force ?

- (1) $d\sqrt{k}$ (2) $k\sqrt{d}$
(3) $1.5d\sqrt{k}$ (4) $2d\sqrt{k}$

Space for your notes:

Q2 - 24 January - Shift 1

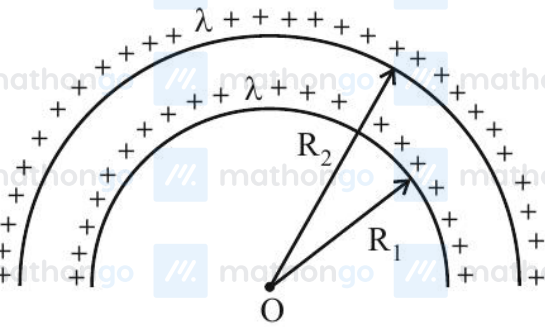
A stream of a positively charged particles having $\frac{q}{m} = 2 \times 10^{11} \frac{C}{kg}$ and velocity $\vec{v}_0 = 3 \times 10^7 \hat{i} \text{ m/s}$ is deflected by an electric field $1.8\hat{j} \text{ kV/m}$. The electric field exists in a region of 10 cm along x direction. Due to the electric field, the deflection of the charge particles in the y direction is _____ mm.

Space for your notes:

Q3 - 24 January - Shift 2

The electric potential at the centre of two concentric half rings of radii R_1 and R_2 , having same linear charge density λ is

Space for your notes:



(1) $\frac{2\lambda}{\epsilon_0}$

(2) $\frac{\lambda}{2\epsilon_0}$

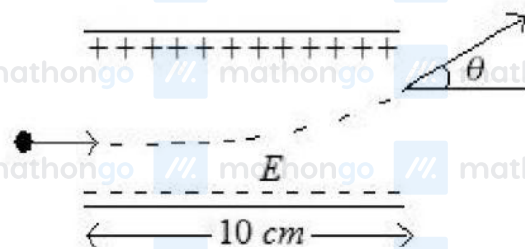
(3) $\frac{\lambda}{4\epsilon_0}$

(4) $\frac{\lambda}{\epsilon_0}$

Q4 - 25 January - Shift 1

A uniform electric field of 10 N/C is created between two parallel charged plates (as shown in figure). An electron enters the field symmetrically between the plates with a kinetic energy 0.5 eV. The length of each plate is 10 cm. The angle (θ) of deviation of the path of electron as it comes out of the field is _____ (in degree).

Space for your notes:



Q5 - 25 January - Shift 2

A point charge of $10\text{ }\mu\text{C}$ is placed at the origin. At what location on the X-axis should a point charge of $40\mu\text{C}$ be placed so that the net electric field is zero at $x = 2\text{ cm}$ on the X-axis ?

Space for your notes:

(1) $x = 6\text{ cm}$

(2) $x = 4\text{ cm}$

(3) $x = 8\text{ cm}$

(4) $x = -4\text{ cm}$

Q6 - 25 January - Shift 2

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Match List I with List II :

A.	Gauss's Law in Electrostatics	I.	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$
B.	Faraday's Law	II.	$\oint \vec{B} \cdot d\vec{A} = 0$
C.	Gauss's Law in Magnetism	III.	$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_C + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$
D.	Ampere-Maxwell Law	IV.	$\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$

Space for your notes:

Choose the correct answer from the options given below :

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-IV, D-I

Q7 - 29 January - Shift 1

In a cuboid of dimension $2L \times 2L \times L$, a charge q is placed at the centre of the surface 'S' having area of $4L^2$. The flux through the opposite surface to 'S' is given by

- (1) $\frac{q}{12\epsilon_0}$
- (2) $\frac{q}{3\epsilon_0}$
- (3) $\frac{q}{2\epsilon_0}$
- (4) $\frac{q}{6\epsilon_0}$

Space for your notes:

Q8 - 29 January - Shift 1

Ratio of thermal energy released in two resistor R and 3R connected in parallel in an electric circuit is :

- (1) 3 : 1 (2) 1 : 1
(3) 1 : 3 (4) 1 : 27

Space for your notes:

Q9 - 29 January - Shift 1

A point charge $q_1 = 4q_0$ is placed at origin.

Another point charge $q_2 = -q_0$ is placed at $x = 12$ cm. Charge of proton is q_0 . The proton is placed on x-axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is _____ cm.

Space for your notes:

Q10 - 29 January - Shift 2

A point charge 2×10^{-2} C is moved from P to S in a uniform electric field of 30 NC^{-1} directed along positive x-axis. If coordinates of P and S are (1, 2, 0) m and (0, 0, 0) m respectively, the work done by electric field will be

- (1) 1200 mJ (2) 600 mJ
(3) -600 mJ (4) -1200 mJ

Space for your notes:

Q11 - 29 January - Shift 2

For a charged spherical ball, electrostatic potential inside the ball varies with r as $V = 2ar^2 + b$.

Here, a and b are constant and r is the distance from the center. The volume charge density inside the ball is $-\lambda\epsilon$. The value of λ is _____.

ϵ = permittivity of medium.

Space for your notes:

Q12 - 30 January - Shift 1

Two isolated metallic solid spheres of radii R and $2R$ are charged such that both have same charge density σ . The spheres are then connected by a thin conducting wire. If the new charge density of the bigger sphere is σ' . The ratio $\frac{\sigma'}{\sigma}$ is :

(1) $\frac{9}{4}$

(2) $\frac{4}{3}$

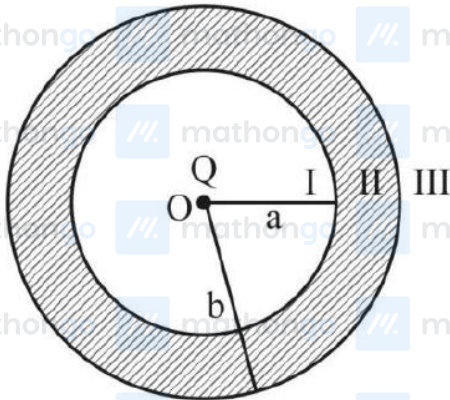
(3) $\frac{5}{3}$

(4) $\frac{5}{6}$

Space for your notes:

Q13 - 30 January - Shift 2

As shown in the figure, a point charge Q is placed at the centre of conducting spherical shell of inner radius a and outer radius b . The electric field due to charge Q in three different regions I, II and III is given by : (I : $r < a$, II : $a < r < b$, III : $r > b$)



Space for your notes:

(1) $E_I = 0, E_{II} = 0, E_{III} \neq 0$

(2) $E_I \neq 0, E_{II} = 0, E_{III} \neq 0$

(3) $E_I \neq 0, E_{II} = 0, E_{III} = 0$

(4) $E_I = 0, E_{II} = 0, E_{III} = 0$

Q14 - 30 January - Shift 2

A point source of 100 W emits light with 5% efficiency. At a distance of 5 m from the source, the intensity produced by the electric field component is :

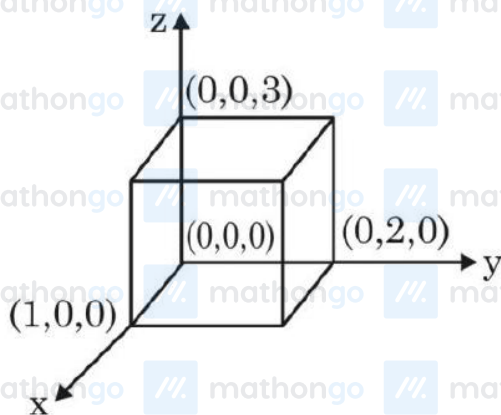
- (1) $\frac{1}{2\pi} \frac{W}{m^2}$ (2) $\frac{1}{40\pi} \frac{W}{m^2}$
 (3) $\frac{1}{10\pi} \frac{W}{m^2}$ (4) $\frac{1}{20\pi} \frac{W}{m^2}$

Space for your notes:

Q15 - 30 January - Shift 2

As shown in figure, a cuboid lies in a region with electric field $\mathbf{E} = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}$ N/C. The magnitude of charge within the cuboid is $n\epsilon_0$ C.

The value of n is _____ (if dimension of cuboid is $1 \times 2 \times 3$ m³)



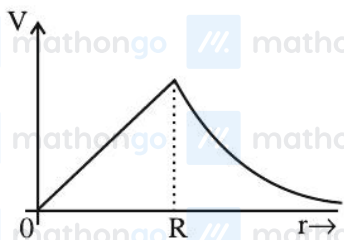
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Q16 - 31 January - Shift 1

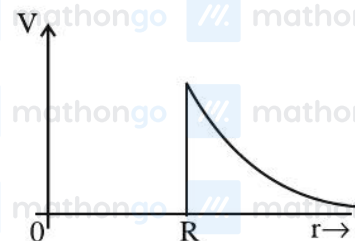
Which of the following correctly represents the variation of electric potential (V) of a charged spherical conductor of radius (R) with radial distance (r) from the centre?

Space for your notes:

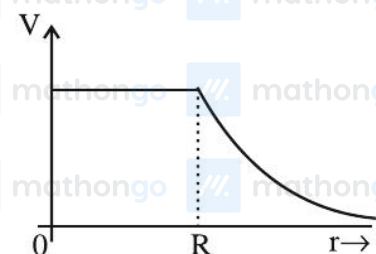
(1)



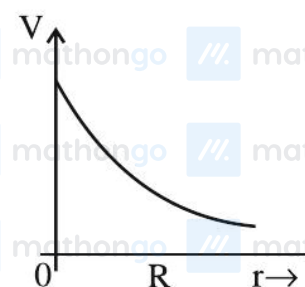
(2)



(3)



(4)



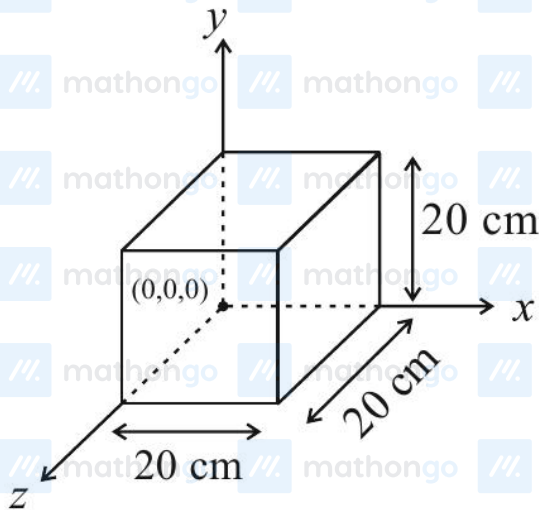
Q17 - 31 January - Shift 1

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Expression for an electric field is given by

$$\vec{E} = 4000x^2\hat{i} \frac{\text{V}}{\text{m}}.$$

The electric flux through the cube of side 20 cm when placed in electric field (as shown in the figure) is _____ V cm.



Space for your notes:

Q18 - 31 January - Shift 2

Considering a group of positive charges, which of the following statements is correct?

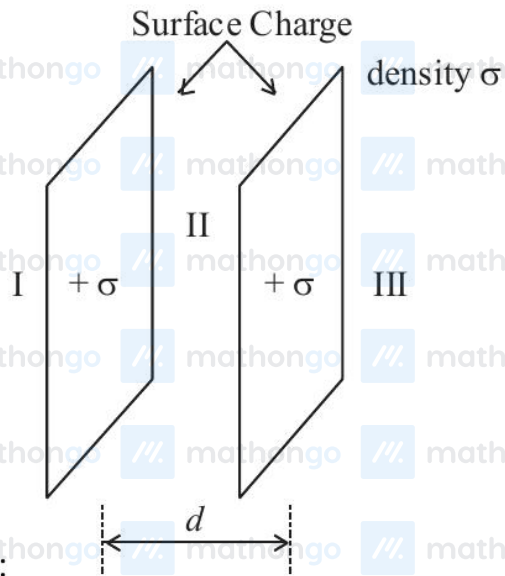
- (1) Net potential of the system cannot be zero at a point but net electric field can be zero at that point.
- (2) Net potential of the system at a point can be zero but net electric field can't be zero at that point.
- (3) Both the net potential and the net field can be zero at a point.
- (4) Both the net potential and the net electric field cannot be zero at a point.

Space for your notes:

Q19 - 01 February - Shift 1

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Let σ be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different region E_I , E_{II} and



E_{III} are:

$$(1) \vec{E}_I = \frac{2\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0} \hat{n}$$

$$(2) \vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{III} = 0$$

$$(3) \vec{E}_I = \frac{\sigma}{2\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0} \hat{n}$$

$$(4) \vec{E}_I = -\frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0} \hat{n}$$

Space for your notes:

Q20 - 01 February - Shift 1

Two equal positive point charges are separated by a distance $2a$. The distance of a point from the centre of the line joining two charges on the equatorial line (perpendicular bisector) at which force experienced by a test charge q_0 becomes

maximum is $\frac{a}{\sqrt{x}}$. The value of x is _____.

Space for your notes:

Q21 - 01 February - Shift 2

A cubical volume is bounded by the surfaces $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$. The electric

field in the region is given by $\vec{E} = E_0 x \hat{i}$. Where

$E_0 = 4 \times 10^4 \text{ NC}^{-1} \text{ m}^{-1}$. If $a = 2 \text{ cm}$, the charge contained in the cubical volume is $Q \times 10^{-14} \text{ C}$. The value of Q is ____.

Take $\epsilon_0 = 9 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1)

Q2 (2)

Q3 (2)

Q4 (45)

Q5 (1)

Q6 (1)

Q7 (4)

Q8 (1)

Q9 (24)

Q10 (3)

Q11 (12)

Q12 (4)

Q13 (2)

Q14 (2)

Q15 (12)

Q16 (3)

Q17 (640)

Q18 (1)

Q19 (4)

Q20 (2)

Q21 (288)

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Q1 (1)

$$F = \frac{1}{(4\pi\epsilon_0)} \frac{q_1 q_2}{kd^2} \text{ (in medium)}$$

$$F_{\text{Air}} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{d'^2}$$

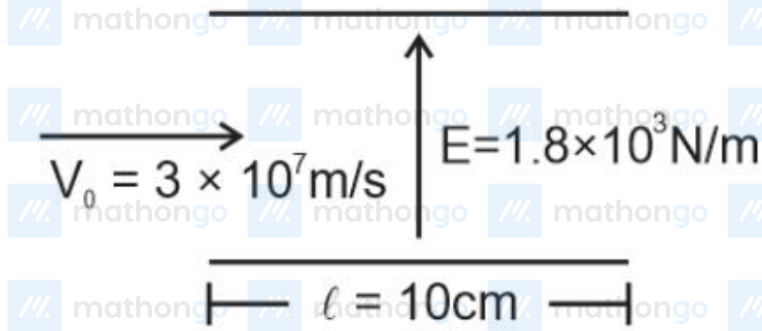
$$F = F_{\text{Air}}$$

$$\frac{q_1 q_2}{4\pi\epsilon_0 kd^2} = \frac{q_1 q_2}{4\pi\epsilon_0 d'^2}$$

$$d' = d\sqrt{k}$$

Q2 (2)

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$$a = \frac{F}{m} = \frac{qE}{m} = (2 \times 10^{11})(1.8 \times 10^3)$$

$$= 3.6 \times 10^{14} \text{ m/s}^2$$

$$\text{Time to cross plates} = \frac{d}{v}$$

$$t = \frac{0.10}{3 \times 10^7}$$

$$y = \frac{1}{2}at^2 = \frac{1}{2}(3.6 \times 10^{14})\left(\frac{0.01}{9 \times 10^{14}}\right)$$

$$= 0.2 \times 0.01$$

$$= 0.002 \text{ m}$$

$$= 2 \text{ mm}$$

Q3 (2)

Potential at centre

$$V = \frac{(\lambda \cdot \pi R_2)}{4\pi\epsilon_0 R_2} + \frac{(\lambda \cdot \pi R_1)}{4\pi\epsilon_0 R_1}$$

$$= \frac{\lambda}{2\epsilon_0}$$

Q4 (45)

$$0.5e = \frac{1}{2}mv_x^2 \Rightarrow v_x = \sqrt{\frac{e}{m}}$$

$$\text{Along x } L = v_x t = \sqrt{\frac{e}{m}} t$$

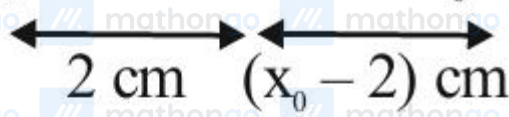
$$\text{Along y } v_y = \frac{eE}{m} t$$

$$\text{dividing } \frac{v_y}{L} = E \sqrt{\frac{e}{m}} = E v_x$$

$$\Rightarrow \tan \theta = \frac{v_y}{v_x} = E \times L = 10 \times 0.1 = 1$$

$$\theta = 45^\circ$$

Q5 (1)



$$E_P = \frac{K \times 10}{2^2} - \frac{K \times 40}{(x_0 - 2)^2} = 0$$

$$\frac{1}{2} = \frac{2}{x_0 - 2}$$

$$x_0 - 2 = 4$$

$$x_0 = 6 \text{ cm}$$

Q6 (1)

Gauss's Law of electrostatic

$$\phi = \oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

Faraday's law $\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt}$

Gauss's law of magnetism $\oint \vec{B} \cdot d\vec{A} = 0$

Ampere's Maxwell law

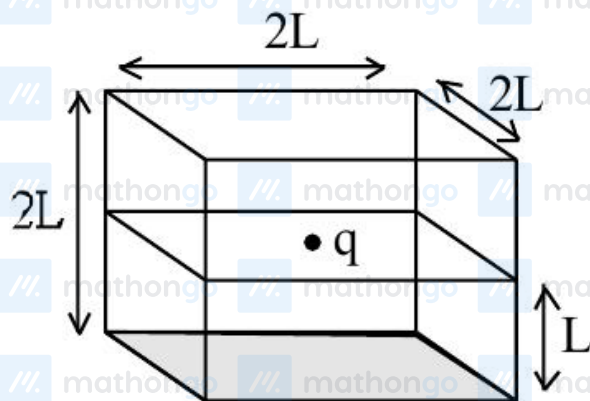
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_C + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

Where i_C : Conduction current

$\epsilon_0 \frac{d\phi_E}{dt}$: Displacement current

Q7 (4)

$$\phi = \frac{Q / \epsilon_0}{6}$$



Flux passing through shaded face = $\frac{q}{6\epsilon_0}$

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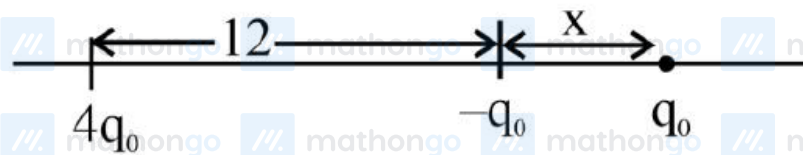
Q8 (1)

$$H = \frac{V^2}{R} \times t$$

$$V^2 t$$

$$\frac{H_1}{H_2} = \frac{\frac{R}{V^2 t}}{\frac{3R}{V^2 t}} = 3:1$$

Q9 (24)

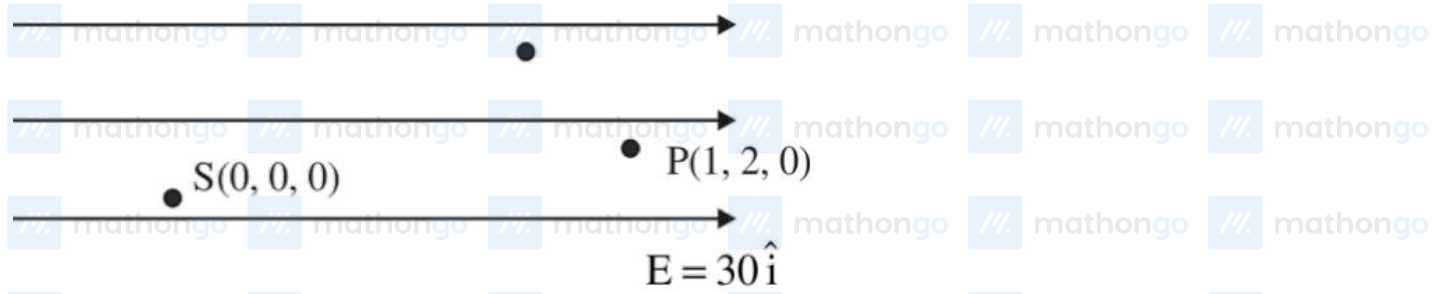


$$\frac{q_0}{x^2} = \frac{4q_0}{(x+12)^2}$$

$$x+12 = 2x$$

$$x = 12$$

Q10 (3)



$$\begin{aligned}\omega_E &= q \vec{E} \cdot \vec{S} \\ &= 2 \times 10^{-2} [30\hat{i} \cdot (-\hat{i})] \\ &= 2 \times 10^{-2} (-30) \\ &= -60 \times 10^{-2} \\ &= -\frac{60}{100} = -0.6 \text{ J} \\ &= -600 \text{ mJ}\end{aligned}$$

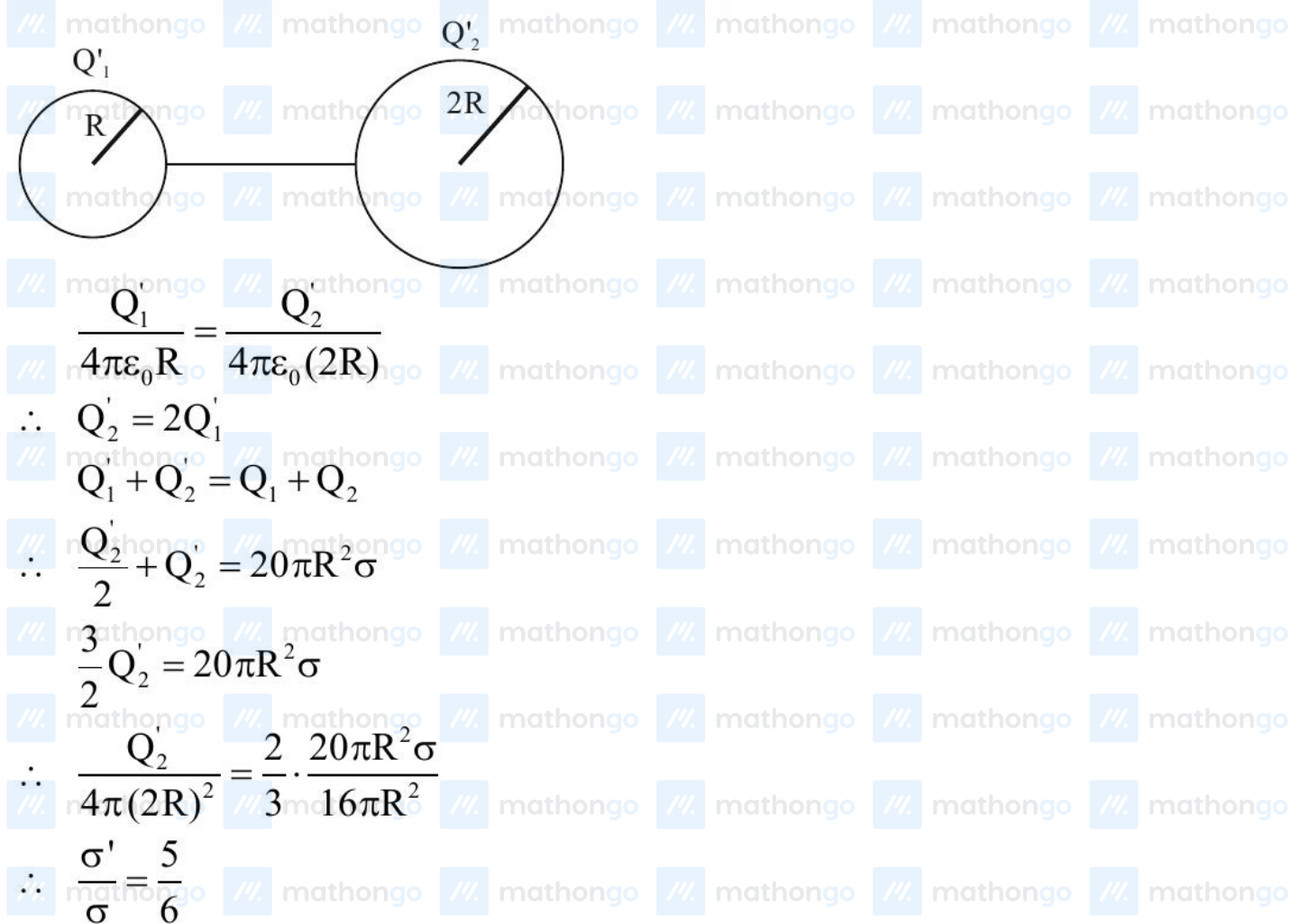
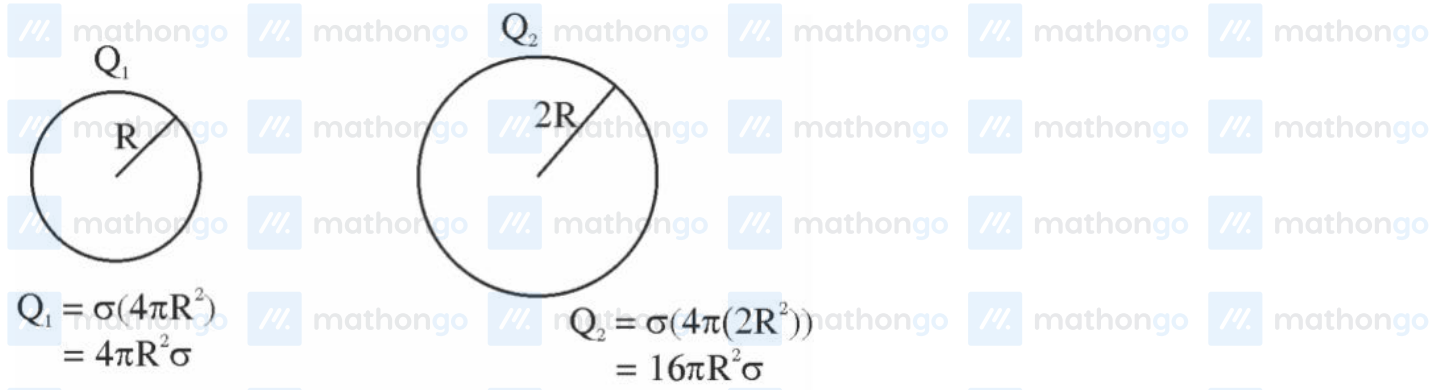
Q11 (12)

$$E = -\frac{dV}{dr} = -4ar \equiv \frac{\rho r}{3\epsilon_0} \text{ (compare)}$$

Result inside uniformly charged solid sphere

$$\begin{aligned}\rho &= -12a\epsilon_0 \\ \lambda &= 12\end{aligned}$$

Q12 (4)

**Q13 (2)**

Electric field inside material of conductor is zero.

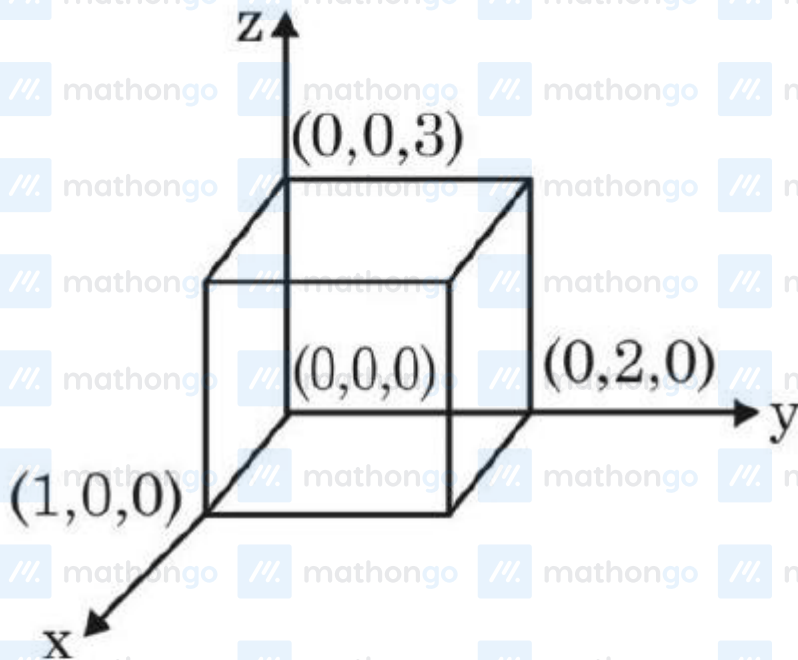
Q14 (2)

$$I_{EF} = \frac{1}{2} \times \frac{5}{4\pi \times 5^2}$$

$$= \frac{1}{40\pi} \text{ W/m}^2$$

Q15 (12)

$$\vec{E} = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}$$



$$\phi_{\text{net}} = -8 \times 3 + 2 \times 6 = -12$$

$$-12 = \frac{q}{\epsilon_0}$$

$$|q| = 12\epsilon_0$$

Q16 (3)

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Conceptual

Q17 (640)

$$\text{Flux} = \vec{E} \cdot \vec{A}$$

$$= 4000 (0.2)^2 \frac{\text{V}}{\text{m}} \cdot (0.2)^2 \text{m}^2$$

$$= 4000 \times 16 \times 10^{-4} \text{Vm}$$

$$= 640 \text{ Vcm}$$

Ans. 640

Q18 (1)

No Solution

Q19 (4)

Assuming RHS to be $\hat{n}$

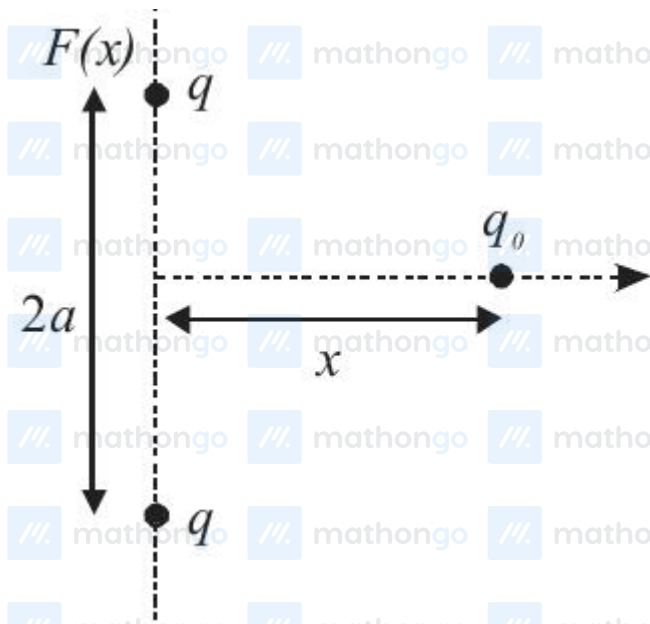
$$\vec{E}_I = \frac{\sigma}{2\epsilon_0}(-\hat{n}) + \frac{\sigma}{2\epsilon_0}(-\hat{n}) = -\frac{\sigma}{\epsilon_0}\hat{n}$$

$$\vec{E}_{II} = 0,$$

$$\vec{E}_{III} = \frac{\sigma}{2\epsilon_0}(\hat{n}) + \frac{\sigma}{2\epsilon_0}(\hat{n}) = \frac{\sigma}{\epsilon_0}(\hat{n})$$

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Q20 (2)



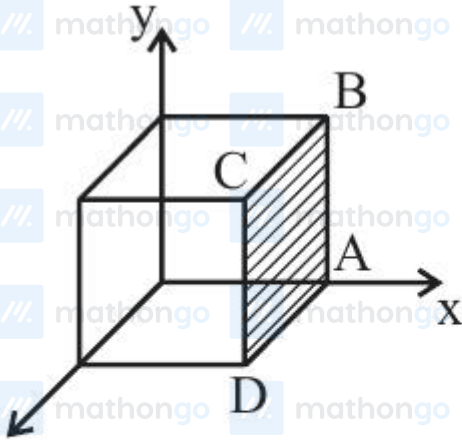
$$F = \frac{2Kqq_0x}{(x^2 + a^2)^{3/2}}$$

For F to be maximum

$$\frac{dF}{dx} = 0$$

$$x = \frac{a}{\sqrt{2}}$$

Q21 (288)



$$\vec{E} = E_0 x \hat{i}$$

$$\phi_{\text{net}} = \phi_{\text{ABCD}} = E_0 a \cdot a^2$$

$$\frac{q_{\text{en}}}{\epsilon_0} = E_0 a^3$$

$$q_{\text{en}} = E_0 \epsilon_0 a^3$$

$$= 4 \times 10^4 \times 9 \times 10^{-12} \times 8 \times 10^{-6}$$

$$= 288 \times 10^{-14} \text{ C}$$

$$Q = 288$$

$$\text{Ans. } 288$$